

CBSE Class 12 Maths — Half-Yearly Predicted Paper (Set 4 of 10)

Based on the highly analytical and competency-based pattern of the most recent CBSE board papers and official sample question papers, here is Set 4 of the Half-Yearly Predicted Paper. This set exclusively covers the ~80% half-yearly syllabus and excludes Linear Programming and Probability entirely. Every question is uniquely selected from high-weightage frameworks.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Let $A = \{1, 2, 3\}$. Find the total number of relations on set A which are reflexive. * **Chapter:** Relations and Functions * **Confidence: High**

Q2. Find the principal value of $\cot\left(\frac{\pi}{2} - 2\cot^{-1}\sqrt{3}\right)$. * **Chapter:** Inverse Trigonometric Functions * **Confidence: High**

Q3. If A is a square matrix of order 3 with $|A| = 4$, find the exact value of $|2A^{-1}|$. * **Chapter:** Matrices * **Confidence: Very High**

Q4. Find the maximum value of the determinant $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 + \sin \theta & 1 \\ 1 & 1 & 1 + \cos \theta \end{vmatrix}$. * **Chapter:** Determinants * **Confidence: Medium**

Q5. Determine the value of k if the function $f(x) = \begin{cases} \frac{\sin 3x}{5x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$. * **Chapter:** Continuity and Differentiability * **Confidence: Very High**

Q6. What is the derivative of $\log_e(\sec x + \tan x)$ with respect to x ? * **Chapter:** Continuity and Differentiability * **Confidence: High**

Q7. The rate of change of the volume of a sphere is equal to the rate of change of its radius. Find its radius. * **Chapter:** Application of Derivatives * **Confidence: Medium**

Q8. Evaluate the indefinite integral: $\int \frac{e^x(1+x)}{\cos^2(xe^x)} dx$. * **Chapter:** Integrals * **Confidence: Very High**

Q9. Evaluate the definite integral: $\int_{-1}^1 |x| dx$. * **Chapter:** Integrals * **Confidence: Very High**

Q10. Write the integrating factor (I.F.) for the linear differential equation: $x \frac{dy}{dx} - y = 2x^2$. * **Chapter:** Differential Equations * **Confidence: High**

Q11. If $\vec{a}$ and $\vec{b}$ are unit vectors and $\vec{a} + \vec{b}$ is also a unit vector, find the angle between $\vec{a}$ and $\vec{b}$. * **Chapter:** Vector Algebra * **Confidence: Very High**

Q12. Find the value of λ if the vectors $2\hat{i} + 3\hat{j} - \hat{k}$ and $4\hat{i} + 6\hat{j} + \lambda\hat{k}$ are parallel to each other. * **Chapter:** Vector Algebra * **Confidence: High**

Q13. Write the equation of the x-axis in Cartesian form. * **Chapter:** Three-Dimensional Geometry * **Confidence: Medium**

Q14. Find the perpendicular distance of the point $(3, 4, 5)$ from the xz-plane. * **Chapter:** Three-Dimensional Geometry * **Confidence: High**

Q15. If $A = \begin{bmatrix} 0 & 2 \\ 3 & -4 \end{bmatrix}$ and $kA = \begin{bmatrix} 0 & 3a \\ 2b & 24 \end{bmatrix}$, find the values of k , a , and b . * **Chapter:** Matrices * **Confidence: High**

Q16. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x - 4$. Find the composition function $f(f(x))$. * **Chapter:** Relations and Functions * **Confidence: High**

Q17. Find the absolute minimum value of the function $f(x) = |x + 3| - 1$. * **Chapter:** Application of Derivatives * **Confidence: Medium**

Q18. Evaluate the indefinite integral: $\int \frac{\sin x - \cos x}{\sin x + \cos x} dx$. * **Chapter:** Integrals * **Confidence: Very High**

Q19. (Assertion-Reason): Assertion (A): The points with position vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ are collinear if $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = \vec{0}$. **Reason (R):** The vector area of the triangle formed by these three points is given by $\frac{1}{2}(\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})$. Choose the correct option. * **Chapter:** Vector Algebra * **Confidence: High**

Q20. (Assertion-Reason): Assertion (A): The degree of the differential equation $e^y + y'' = 0$ is not defined. **Reason (R):** The degree of a differential equation is defined only if the equation can be expressed as a polynomial equation in its derivatives. Choose the correct option. * **Chapter:** Differential Equations * **Confidence: Very High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Express $\sin^{-1}\left(\frac{\sin x + \cos x}{\sqrt{2}}\right)$, where $-\frac{\pi}{4} < x < \frac{\pi}{4}$, in its simplest form. * **Chapter:** Inverse Trigonometric Functions * **Confidence: High**

Q22. Find matrix X if $2X + 3 \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 1 & 4 \end{bmatrix}$. * **Chapter:** Matrices * **Confidence:** High

Q23. If $y = \log_e(\sqrt{x} + \sqrt{x-a})$, prove that $\frac{dy}{dx} = \frac{1}{2\sqrt{x(x-a)}}$. * **Chapter:** Continuity and Differentiability * **Confidence:** Medium

Q24. Find the intervals in which the function $f(x) = x^2 - 4x + 6$ is strictly increasing or strictly decreasing. * **Chapter:** Application of Derivatives * **Confidence:** Very High

Q25. Find the angle between the line $\frac{x-2}{3} = \frac{y+1}{-1} = \frac{z-3}{2}$ and the plane $3x + 4y + z + 5 = 0$. * **Chapter:** Three-Dimensional Geometry * **Confidence:** Very High

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Show that the relation R defined in the set A of all polygons as $R = \{(P_1, P_2) : P_1 \text{ and } P_2 \text{ have the same number of sides}\}$ is an equivalence relation. * **Chapter:** Relations and Functions * **Confidence:** High

Q27. Using properties of determinants, solve for x : $\begin{vmatrix} x+a & b & c \\ a & x+b & c \\ a & b & x+c \end{vmatrix} = 0$. * **Chapter:** Determinants * **Confidence:** Very High

Q28. If $x = a(\cos t + \log \tan \frac{t}{2})$ and $y = a \sin t$, find the value of $\frac{dy}{dx}$. * **Chapter:** Continuity and Differentiability * **Confidence:** Very High

Q29. Evaluate the indefinite integral: $\int \frac{1}{\cos(x-a)\cos(x-b)} dx$. * **Chapter:** Integrals * **Confidence:** High

Q30. Find the particular solution of the differential equation $\frac{dy}{dx} + y \cot x = 4x \operatorname{cosec} x$, given that $y = 0$ when $x = \frac{\pi}{2}$. * **Chapter:** Differential Equations * **Confidence:** Very High

Q31. If $\vec{a}$, $\vec{b}$, and $\vec{c}$ are mutually perpendicular vectors of equal magnitudes, show that the vector $\vec{a} + \vec{b} + \vec{c}$ is equally inclined to $\vec{a}$, $\vec{b}$, and $\vec{c}$. * **Chapter:** Vector Algebra * **Confidence:** Very High

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Given $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$, find A^{-1} . Use A^{-1} to solve the following system of equations: $2x - 3y + 5z = 11$

$3x + 2y - 4z = -5$ $x + y - 2z = -3$ * **Chapter:** Matrices and Determinants * **Confidence:** Very High

Q33. An open box with a square base is to be made out of a given quantity of cardboard of area c^2 . Show mathematically that the maximum volume of the box is $\frac{c^3}{6\sqrt{3}}$. * **Chapter:** Application of Derivatives * **Confidence:** Very High

Q34. Evaluate the following definite integral: $\int_0^{\pi/4} \frac{\sin x + \cos x}{9 + 16 \sin 2x} dx$. * **Chapter:** Integrals * **Confidence:** High

Q35. Find the coordinates of the foot of the perpendicular and the exact length of the perpendicular drawn from the point $P(2, 4, -1)$ to the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$. * **Chapter:** Three-Dimensional Geometry * **Confidence:** Very High

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Application of Integrals (Land Boundary) A farmer has a piece of land shaped like the region bounded by the curve $y = 2\sqrt{x}$, the line $x = 4$, and the x-axis in the first quadrant. He wishes to cultivate this specific region and needs to calculate its exact area to determine the seeds required. (i) Express the total area of the land as a definite integral. (1 Mark) (ii) Find the exact total area of the land. (1 Mark) (iii) If the farmer wants to divide the land into two equal areas using a vertical fence along the line $x = k$, find the exact value of k . (2 Marks) * **Chapter:** Application of Integrals * **Confidence:** High

Q37. Case Study 2: Differential Equations (Bacterial Growth) In a laboratory culture, the initial bacteria count is observed to be 100,000. It is noted that the number of bacteria is increasing at a rate proportional to the number present at that instant. Let $N(t)$ be the bacteria count at time t hours. (i) Write the differential equation governing this bacterial growth. (1 Mark) (ii) If the number of bacteria is increased by 10% in exactly 2 hours, find the general solution expressing $N(t)$ in terms of t and the constant of proportionality k . (1 Mark) (iii) Find the exact time t (expressed in logarithms) when the bacteria count will reach 200,000. (2 Marks) * **Chapter:** Differential Equations * **Confidence:** Very High

Q38. Case Study 3: Vector Algebra (Force and Work) A team of workers is moving a heavy industrial crate. Two independent constant forces, $\vec{F}_1 = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{F}_2 = \hat{i} - 2\hat{j} + 4\hat{k}$ (measured in Newtons), act simultaneously on the crate, displacing it in a straight line from point $A(1, 2, 3)$ to point $B(3, 4, 5)$ (measured in meters). (i) Find the resultant force

$\vec{F}$ acting on the crate. (1 Mark) (ii) Find the displacement vector $\vec{AB}$. (1 Mark) (iii) Calculate the total work done by the forces in displacing the crate. (2 Marks) * **Chapter:** Vector Algebra * **Confidence:** High

Answer Key

Q1. Number of reflexive relations = $2^{n(n-1)} = 2^{3(2)} = 2^6 = 64$ **Q2.**

$\cot\left(\frac{\pi}{2} - 2\cot^{-1}\sqrt{3}\right) = \tan(2\cot^{-1}\sqrt{3}) = \tan\left(2 \times \frac{\pi}{6}\right) = \tan\left(\frac{\pi}{3}\right) = \sqrt{3}$. **Q3.** $|2A^{-1}| = 2^3|A^{-1}| = 8 \times \frac{1}{|A|} = \frac{8}{4} = 2$.

Q4. Using $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$, $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 0 & \sin\theta & 0 \\ 0 & 0 & \cos\theta \end{vmatrix} = \sin\theta \cos\theta = \frac{1}{2}\sin 2\theta$. Max value of $\sin 2\theta$ is 1, so max $\Delta = \frac{1}{2}$. **Q5.** $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x} = \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{3}{5} = 1 \times \frac{3}{5} = \frac{3}{5}$. So, $k = \frac{3}{5}$. **Q6.**

$y' = \frac{1}{\sec x + \tan x} \cdot (\sec x \tan x + \sec^2 x) = \frac{\sec x(\tan x + \sec x)}{\sec x + \tan x} = \sec x$. **Q7.** $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. Given $\frac{dV}{dt} = \frac{dr}{dt} \Rightarrow 4\pi r^2 = 1 \Rightarrow r = \frac{1}{\sqrt{4\pi}}$. **Q8.** Let $t = xe^x \Rightarrow dt = (e^x + xe^x)dx = e^x(1+x)dx$. Integral becomes $\int \frac{dt}{\cos^2 t} = \int \sec^2 t dt = \tan t + C = \tan(xe^x) + C$. **Q9.**

$\int_{-1}^1 |x| dx = \int_{-1}^0 (-x) dx + \int_0^1 x dx = \left[-\frac{x^2}{2}\right]_{-1}^0 + \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2} + \frac{1}{2} = 1$. **Q10.** Divide by x : $\frac{dy}{dx} - \frac{1}{x}y = 2x$. I.F. = $e^{\int (-1/x) dx} = e^{-\log x} = \frac{1}{x}$. **Q11.**

$|\vec{a} + \vec{b}|^2 = 1 \Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} = 1 \Rightarrow 1 + 1 + 2(1)(1)\cos\theta = 1 \Rightarrow 2\cos\theta = -1 \Rightarrow \cos\theta = -1/2 \Rightarrow \theta =$ (or $2\pi/3$). **Q12.** Vectors are proportional: $\frac{2}{4} = \frac{3}{6} = \frac{-1}{\lambda} \Rightarrow \frac{1}{2} = \frac{-1}{\lambda} \Rightarrow \lambda = -2$. **Q13.** $\frac{x-0}{1} = \frac{y-0}{0} = \frac{z-0}{0}$ or simply $\frac{x}{1} = \frac{y}{0} = \frac{z}{0}$. **Q14.** Distance from the xz-plane (where $y = 0$) is simply the $|y|$ -coordinate, which is 4 units. **Q15.**

$kA = \begin{bmatrix} 0 & 2k \\ 3k & -4k \end{bmatrix} = \begin{bmatrix} 0 & 3a \\ 2b & 2a \end{bmatrix}$. Thus, $-4k = 2a \Rightarrow k = -a/2$. $3a = 2b \Rightarrow b = 3a/2$. $2b = 3k \Rightarrow 3a = -3a/2 \Rightarrow a = 0$. **Q16.** $f(f(x)) = f(3x-4) = 3(3x-4)-4 = 9x-12-4 = 9x-16$. **Q17.** Absolute minimum of $|x+3|$ is 0 (at $x = -3$). Thus, min value of $f(x)$ is $0-1 = -1$. **Q18.** Let

$t = \sin x + \cos x \Rightarrow dt = (\cos x - \sin x)dx = -(\sin x - \cos x)dx$. Integral is $\int \frac{dt}{t} = -\log|t| + C = -\log|\sin x + \cos x| + C$. **Q19.** Option A. Both A and R are true, and R is the correct explanation of A (Area of triangle is 0 means points are collinear). **Q20.** Option A. Both A and R are true, and R is the correct explanation of A.

Q21. $\frac{\sin x + \cos x}{\sqrt{2}} = \sin x \cdot \frac{1}{\sqrt{2}} + \cos x \cdot \frac{1}{\sqrt{2}} = \sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4} = \sin\left(x + \frac{\pi}{4}\right)$. Expression simplifies to $x + \frac{\pi}{4}$.

Q22. $2X = \begin{bmatrix} 5 & -2 \\ 1 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 6 \\ 9 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -8 \\ -8 & 4 \end{bmatrix}$. Thus, $X = \begin{bmatrix} 1 & -4 \\ -4 & 2 \end{bmatrix}$. **Q23.**

$\frac{dy}{dx} = \frac{1}{\sqrt{x} + \sqrt{x-a}} \cdot \left(\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{x-a}}\right) = \frac{1}{\sqrt{x} + \sqrt{x-a}} \cdot \frac{\sqrt{x-a} + \sqrt{x}}{2\sqrt{x}\sqrt{x-a}} = \frac{1}{2\sqrt{x^2-a^2}}$. **Q24.** $f'(x) = 2x - 4$. For strictly increasing, $f'(x) > 0 \Rightarrow x > 2$, interval is $(2, \infty)$. For strictly decreasing, $f'(x) < 0 \Rightarrow x < 2$, interval is $(-\infty, 2)$. **Q25.**

Direction vector $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. Normal to plane $\vec{n} = 3\hat{i} + 4\hat{j} + \hat{k}$. $\sin\theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|} = \frac{|9-4+2|}{\sqrt{9+1+4}\sqrt{9+16+1}} = \frac{7}{\sqrt{14}\sqrt{26}} = \frac{7}{\sqrt{364}}$. Thus, $\theta = \sin^{-1}\left(\frac{7}{\sqrt{364}}\right)$.

Q26. Reflexive: P_1 has the same number of sides as P_1 . Symmetric: If P_1, P_2 share the same side count, then P_2, P_1 do too. Transitive: If P_1, P_2 share the same side count, and P_2, P_3 do, then P_1, P_3 do. It's an equivalence relation. **Q27.** Apply

$C_1 \rightarrow C_1 + C_2 + C_3$. Factor out $(x+a+b+c)$. Det becomes $\begin{vmatrix} 1 & b & c \\ 1 & x+b & c \\ 1 & b & x+c \end{vmatrix}$. $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

yields x^2 . Thus, $x^2(x+a+b+c) = 0 \Rightarrow x = 0$ or $x = -(a+b+c)$. **Q28.**

$\frac{dx}{dt} = a\left(-\sin t + \frac{1}{\tan(t/2)} \cdot \sec^2(t/2) \cdot \frac{1}{2}\right) = a\left(-\sin t + \frac{1}{\sin t}\right) = \frac{a\cos^2 t}{\sin t}$. $\frac{dy}{dt} = a\cos t$. Thus,

$\frac{dy}{dx} = \frac{a\cos t}{(a\cos^2 t)/\sin t} = \tan t$. **Q29.** Multiply and divide by $\sin(a-b)$. Numerator

$\sin(a-b) = \sin((x-b)-(x-a)) = \sin(x-b)\cos(x-a) - \cos(x-b)\sin(x-a)$. Split fractions:

$\frac{1}{\sin(a-b)} \int [\tan(x-b) - \tan(x-a)] dx = \frac{1}{\sin(a-b)} [\log|\sec(x-b)| - \log|\sec(x-a)|] + C$. **Q30.** I.F.

$= e^{\int \cot x dx} = e^{\log(\sin x)} = \sin x$. Solution is $y \sin x = \int 4x \csc x \cdot \sin x dx = \int 4x dx = 2x^2 + C$. At

$x = \pi/2, y = 0 \Rightarrow 0 = 2(\pi^2/4) + C \Rightarrow C = -\pi^2/2$. Solution: $y \sin x = 2x^2 - \frac{\pi^2}{2}$. **Q31.** Let magnitudes be k .

$|\vec{a} + \vec{b} + \vec{c}|^2 = 3k^2 \Rightarrow |\vec{a} + \vec{b} + \vec{c}| = k\sqrt{3}$. $\cos\theta = \frac{\vec{a} \cdot (\vec{a} + \vec{b} + \vec{c})}{|\vec{a}||\vec{a} + \vec{b} + \vec{c}|} = \frac{k^2}{k \cdot k\sqrt{3}} = \frac{1}{\sqrt{3}}$. Same for $\vec{b}$ and $\vec{c}$.

Q32. $|A| = -1$. $A^{-1} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$. The system of equations translates to

$A^T X = B \Rightarrow X = (A^T)^{-1} B = (A^{-1})^T B$. Multiplying transpose of A^{-1} with B gives $X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Thus

$x = 1, y = 2, z = 3$. **Q33.** Let base side be x and height be y . Area $c^2 = x^2 + 4xy \Rightarrow y = \frac{c^2 - x^2}{4x}$. Volume $V = x^2 y = \frac{x(c^2 - x^2)}{4}$. $V'(x) = \frac{c^2 - 3x^2}{4} = 0 \Rightarrow x = \frac{c}{\sqrt{3}}$. $V''(x) < 0$ guarantees max. Max volume

$$= \frac{(c/\sqrt{3})(c^2 - c^2/3)}{4} = \frac{c^3}{6\sqrt{3}}. \text{ Q34. Let } \sin x - \cos x = t. \text{ Then } (\cos x + \sin x)dx = dt. \text{ Also,}$$

$$t^2 = 1 - \sin 2x \implies \sin 2x = 1 - t^2. \text{ When } x = 0, t = -1; x = \pi/4, t = 0. \text{ Integral is}$$

$$\int_{-1}^0 \frac{dt}{9+16(1-t^2)} = \int_{-1}^0 \frac{dt}{25-16t^2} = \frac{1}{16} \int_{-1}^0 \frac{dt}{(5/4)^2 - t^2} = \frac{1}{16} \cdot \frac{1}{2(5/4)} \left[\log \left| \frac{5/4+t}{5/4-t} \right| \right]_{-1}^0. \text{ Evaluates to } \frac{1}{40} \log 9. \text{ Q35. General point}$$

$$\text{on line: } Q(\lambda - 5, 4\lambda - 3, -9\lambda + 6). \text{ DRs of } PQ: \lambda - 7, 4\lambda - 7, -9\lambda + 7.$$

$$PQ \perp \text{line} \implies 1(\lambda - 7) + 4(4\lambda - 7) - 9(-9\lambda + 7) = 0 \implies 98\lambda = 98 \implies \lambda = 1. \text{ Foot } Q \text{ is } (-4, 1, -3). \text{ Length}$$

$$PQ = \sqrt{(-4-2)^2 + (1-4)^2 + (-3+1)^2} = \sqrt{36+9+4} = \sqrt{49} = 7 \text{ units.}$$

$$\text{Q36. (i) } A = \int_0^4 2\sqrt{x} dx. \text{ (ii) } A = 2 \left[\frac{x^{3/2}}{3/2} \right]_0^4 = \frac{4}{3} (4^{3/2}) = \frac{32}{3} \text{ sq units. (iii)}$$

$$\int_0^k 2\sqrt{x} dx = \frac{16}{3} \implies \frac{4}{3} k^{3/2} = \frac{16}{3} \implies k^{3/2} = 4 \implies k = 4^{2/3} = 16^{1/3} \text{ (or } \approx 2.52). \text{ Q37. (i) } \frac{dN}{dt} = kN. \text{ (ii)}$$

$$\frac{dN}{N} = kdt \implies \log N = kt + C \implies N(t) = N_0 e^{kt}.$$

$$N(2) = 110,000 \implies 1.1N_0 = N_0 e^{2k} \implies e^{2k} = 1.1 \implies k = \frac{1}{2} \log_e(1.1). \text{ (iii)}$$

$$200,000 = 100,000 e^{kt} \implies e^{kt} = 2 \implies t = \frac{\log_e 2}{k} = \frac{2 \log_e 2}{\log_e 1.1} \text{ hours. Q38. (i)}$$

$$\vec{F} = \vec{F}_1 + \vec{F}_2 = (2+1)\hat{i} + (3-2)\hat{j} + (-1+4)\hat{k} = 3\hat{i} + \hat{j} + 3\hat{k}. \text{ (ii)}$$

$$\vec{AB} = (3-1)\hat{i} + (4-2)\hat{j} + (5-3)\hat{k} = 2\hat{i} + 2\hat{j} + 2\hat{k}. \text{ (iii) Work Done}$$

$$W = \vec{F} \cdot \vec{AB} = 3(2) + 1(2) + 3(2) = 6 + 2 + 6 = 14 \text{ Joules.}$$